TIME SERIES ANALYSIS

Stock Price Dataset

Exploratory Analysis, Volatility, Seasonal Decomposition and Moving Average Smoothing

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Data Analytics Internship Project

Contents

Executive Summary

This report presents a time-series analysis of historical stock-price data covering 505 stock symbols and 497,472 observations. The dataset contains the stock symbol, date, opening price, high price, low price, closing price and trading volume. Five symbols with the highest number of complete observations were selected for comparative analysis: AAL, NTRS, NUE, NVDA and PRGO.

The analysis first compared closing-price trajectories and cumulative percentage growth across the five selected companies. NVDA recorded the strongest cumulative growth, while PRGO recorded the weakest growth and incurred losses over the study period. Daily-return volatility was then calculated to examine short-term fluctuations. NVDA had the highest volatility at 2.37%, whereas NTRS had the lowest at 1.38%.

Because NVDA and PRGO represented the strongest and weakest cumulative growth outcomes respectively, they were selected for deeper time-series analysis. Additive seasonal decomposition using a 252-trading-day period showed that long-term trend dominated both stocks, while the seasonal component was comparatively small. NVDA exhibited a strong and accelerating upward trend; PRGO experienced early growth followed by a prolonged decline and later stabilization.

Finally, 5-day and 21-day moving averages were applied to smooth daily closing prices. The 5-day average was more responsive to short-term movements, while the 21-day average provided a smoother view of the medium-term direction. Together, decomposition and moving-average smoothing reinforced the contrasting behaviour of NVDA and PRGO.

1. Introduction

Time-series analysis is useful for studying observations recorded sequentially over time and for separating underlying trends from short-term fluctuations. In financial markets, historical stock prices can be examined to understand changes in price behaviour, volatility and recurring patterns. This project applies time-series methods to a multi-company stock-price dataset, with particular attention to trend, seasonality, volatility and smoothing.

1.1 Project Objectives

- Explore changes in stock closing prices over time.
- Compare cumulative percentage growth among selected stocks.
- Measure and compare daily-return volatility.
- Decompose selected stock-price series into trend, seasonal and residual components.
- Apply moving-average smoothing using 5-day (weekly) and 21-day (monthly) windows.
- Compare the time-series behaviour of the strongest and weakest cumulative-growth stocks.

2. Dataset Overview

The supplied analysis shows that the dataset contains 497,472 rows and seven columns. The date field was initially stored as an object and was later converted to datetime for time-series operations. The dataset contains 505 unique stock symbols.

Variable	Description	Data Type	Non-null observations
symbol	Stock ticker/symbol	object	497,472
date	Trading date	object → datetime	497,472
open	Opening price	float64	497,461
high	Highest price	float64	497,464
low	Lowest price	float64	497,464
close	Closing price	float64	497,472
volume	Trading volume	int64	497,472

The descriptive statistics indicate substantial variation in stock prices and trading volume. The closing price has a mean of approximately \$86.37 and a median of approximately \$64.98, while the maximum recorded closing price is \$2,049.00. Trading volume ranges from zero to 618,237,600 observations in the supplied summary.

3. Selection of Stocks for Detailed Analysis

Rather than selecting a single familiar company arbitrarily, the analysis identified the symbols with the maximum number of observations. Five symbols shared the maximum count of 1,007 observations each: AAL, PRGO, NVDA, NUE and NTRS. These five stocks therefore formed the comparison set.

Symbol	Observations	Role in analysis
AAL	1,007	Comparative benchmark
NTRS	1,007	Comparative benchmark
NUE	1,007	Comparative benchmark
NVDA	1,007	Highest cumulative growth; detailed analysis
PRGO	1,007	Weakest cumulative growth; detailed analysis

4. Exploratory Time-Series Analysis

4.1 Closing Price Behaviour

The closing-price comparison was used as an initial visual assessment of how the five selected companies changed throughout the study period. Because the stocks have different absolute price levels, the raw closing price chart is useful for observing individual trajectories, but percentage growth provides a more meaningful basis for comparing performance.

4.2 Percentage Growth Analysis

Cumulative percentage growth was calculated relative to each company's first observed closing price using the relationship: (current closing price / first closing price − 1) × 100. The resulting series showed a very strong divergence among the selected stocks.

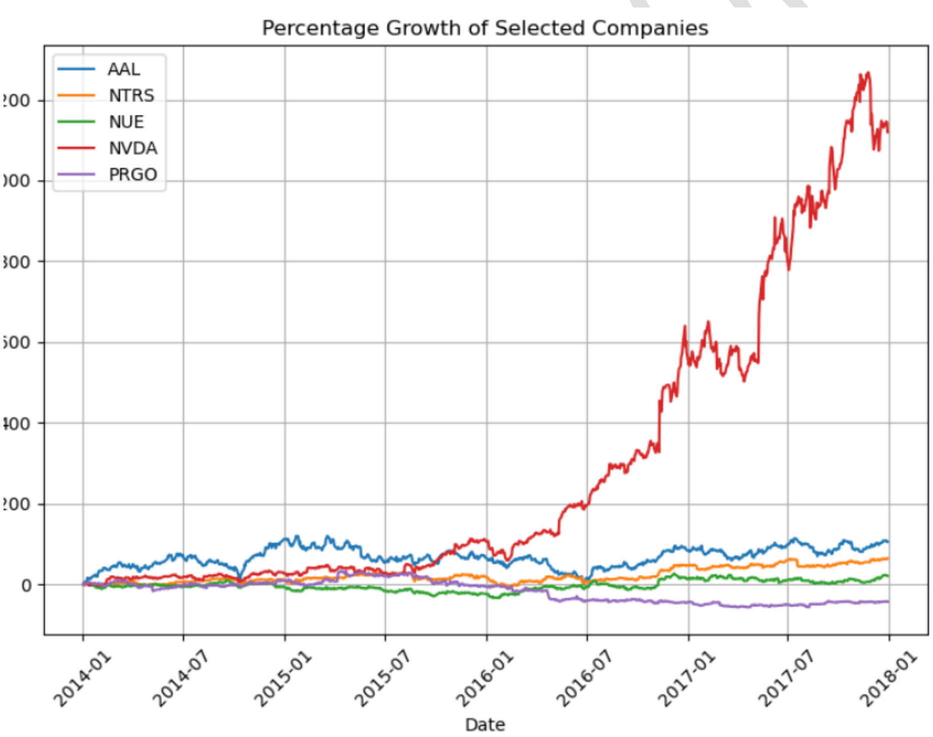

Figure 1. Cumulative percentage growth of the five selected stocks (source analysis).

- NVDA produced by far the strongest cumulative growth, reaching above 1,200% at its peak in the supplied chart.
- PRGO was the weakest performer and remained below its starting level for much of the latter part of the study period.
- The other three stocks displayed positive but considerably smaller cumulative growth relative to NVDA.
- The contrast between NVDA and PRGO provided a clear basis for selecting them for deeper analysis.

5. Daily Return Volatility

Daily return was calculated as the percentage change in closing price from one trading day to the next. The standard deviation of daily returns was then used as the volatility measure. A larger value indicates larger typical day-to-day fluctuations in the closing price series.

Rank	Symbol	Daily-return volatility (%)
1	NTRS	1.38
2	NUE	1.67
3	PRGO	2.14
4	AAL	2.24
5	NVDA	2.37

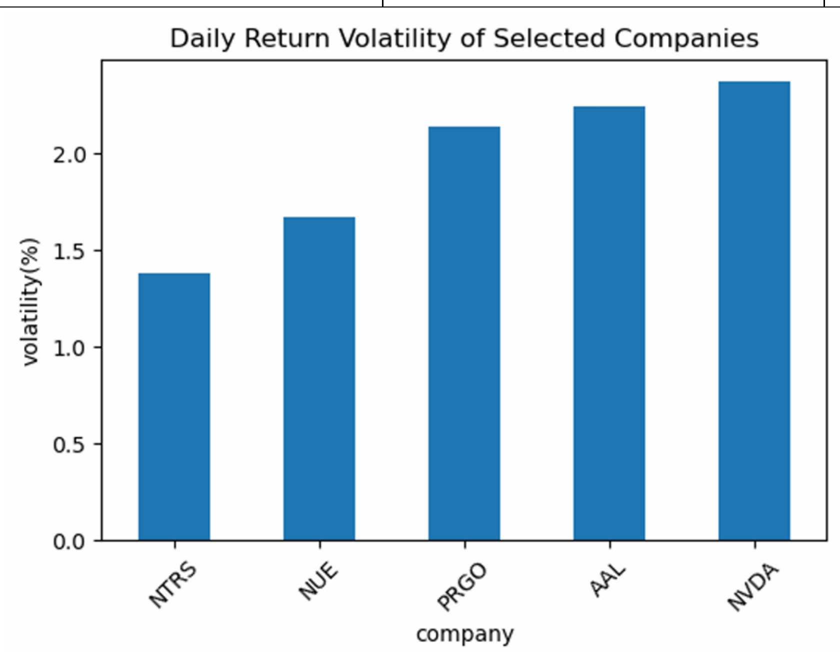

Figure 2. Daily-return volatility of the five selected stocks.

NVDA recorded the highest daily-return volatility at 2.37%, indicating the largest short-term fluctuations among the five stocks. NTRS recorded the lowest volatility at 1.38%, suggesting the most stable daily price movements within this comparison set. PRGO combined relatively high volatility (2.14%) with the weakest long-term growth, meaning that its investors experienced substantial daily fluctuations without a comparable cumulative gain.

The results illustrate the risk–return contrast visible in the selected stocks: the strongest cumulative growth in this sample was accompanied by the highest measured daily volatility. This does not establish a general causal rule, but it is an important observation from this dataset.

6. Seasonal Decomposition

Seasonal decomposition was applied separately to NVDA and PRGO using an additive model and a period of 252 trading days, representing an assumed annual trading cycle. The decomposition separates the observed series into trend, seasonal and residual components.

6.1 NVDA Decomposition

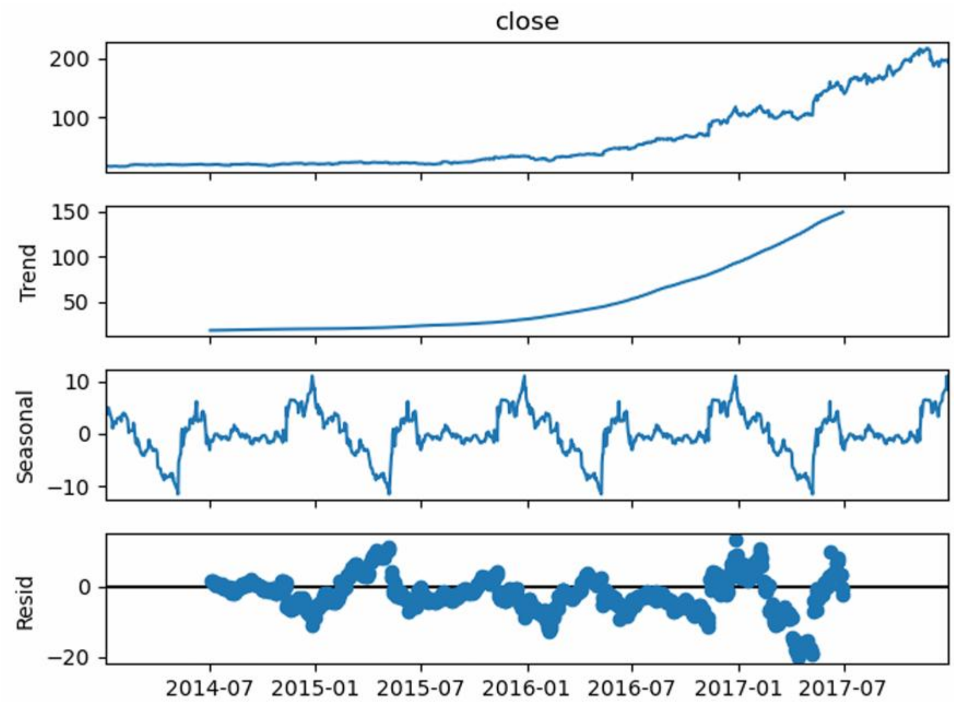

Figure 3. Additive seasonal decomposition of NVDA closing prices.

NVDA shows a relatively slow increase at the beginning of the study period followed by a sustained and accelerating upward trend, particularly from 2016 onward. The seasonal component contains recurring fluctuations under the assumed 252-day annual cycle, but its magnitude is small relative to the overall price movement. The residual component is centred around zero with several notable spikes, indicating irregular movements that are not explained by the trend or seasonal component.

Overall, the decomposition indicates that NVDA's performance during the study period was driven predominantly by long-term trend rather than seasonality, while irregular market movements remained present.

6.2 PRGO Decomposition

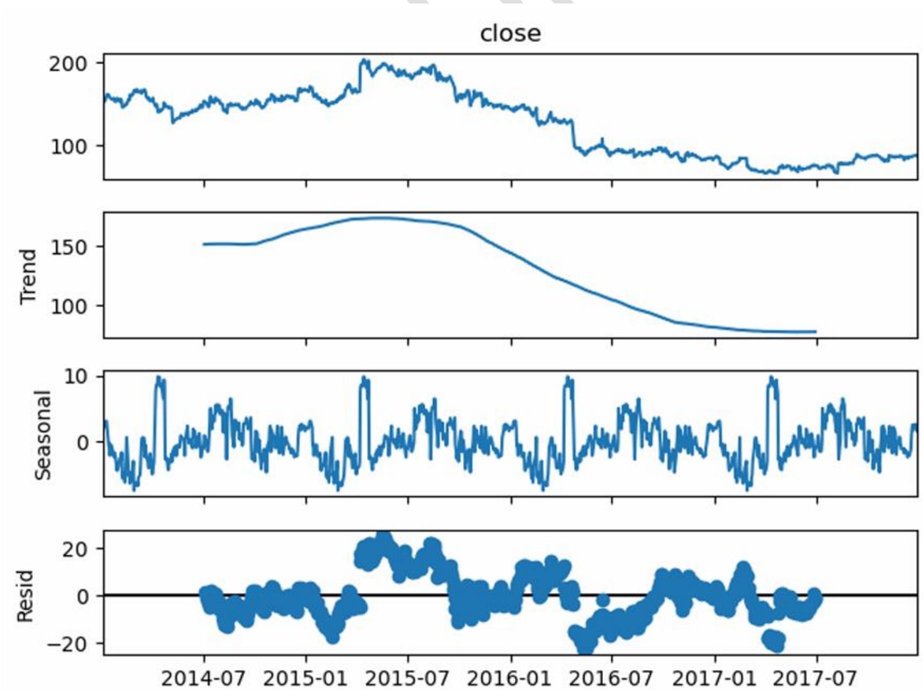

Figure 4. Additive seasonal decomposition of PRGO closing prices.

PRGO began the study period at a relatively high price of approximately \$150 and experienced an upward movement that culminated around 2015. From late 2015 into 2016, the stock entered a prolonged decline. During 2017 the trend flattened and the series showed signs of stabilization and some recovery.

The seasonal component appears more irregular than that of NVDA, but its magnitude remains relatively small compared with the overall long-term decline. The residual component contains irregular movements and breaks, reinforcing the conclusion that seasonality was not the primary driver of PRGO's performance.

6.3 Comparison of Decomposition Results

The two stocks demonstrate strongly contrasting time-series behaviour. NVDA exhibits a sustained and accelerating upward trend, whereas PRGO shows an initial growth phase followed by a prolonged decline. Both series contain recurring seasonal movements under the assumed annual cycle, but these movements are minor relative to their long-term trends. In both cases, residuals capture irregular movements associated with unpredictable events.

7. Moving Average Smoothing

Moving averages were used to reduce short-term noise and make the underlying direction of each stock easier to observe. Two windows were used: a 5-day moving average and a 21-day moving average. The 5-day average responds more quickly to recent changes, while the 21-day average provides a smoother medium-term signal.

7.1 NVDA Moving Average Smoothing

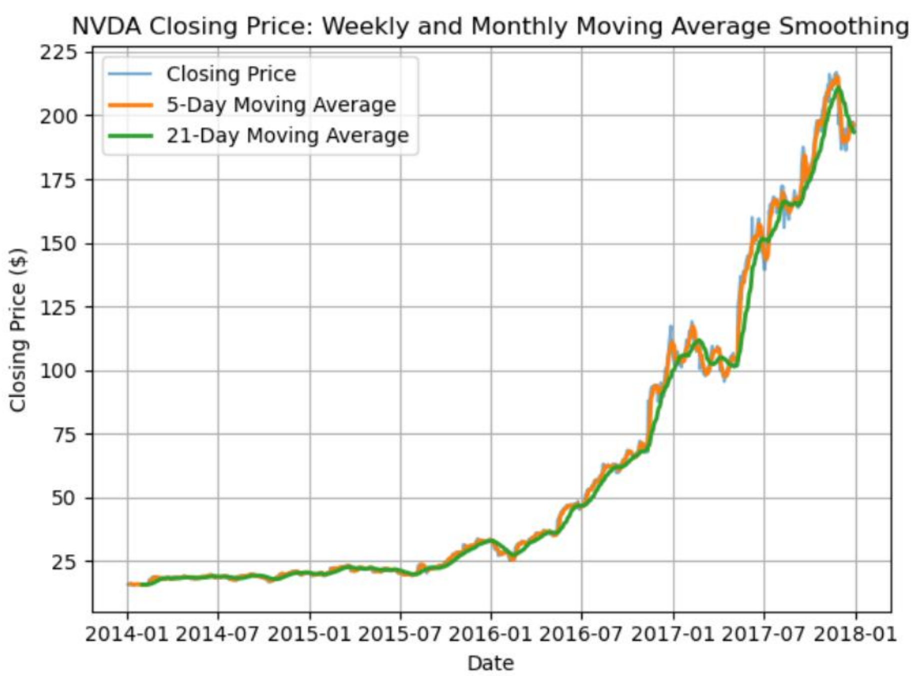

Figure 6. NVDA closing price with 5-day and 21-day moving averages.

The original NVDA closing-price series contains frequent short-term fluctuations. The 5-day moving average closely follows these movements while reducing minor day-to-day noise, making it useful for observing short-term direction. The 21-day moving average is visibly smoother and responds more slowly to abrupt price changes. It therefore provides a clearer representation of the medium-term upward trajectory.

7.2 PRGO Moving Average Smoothing

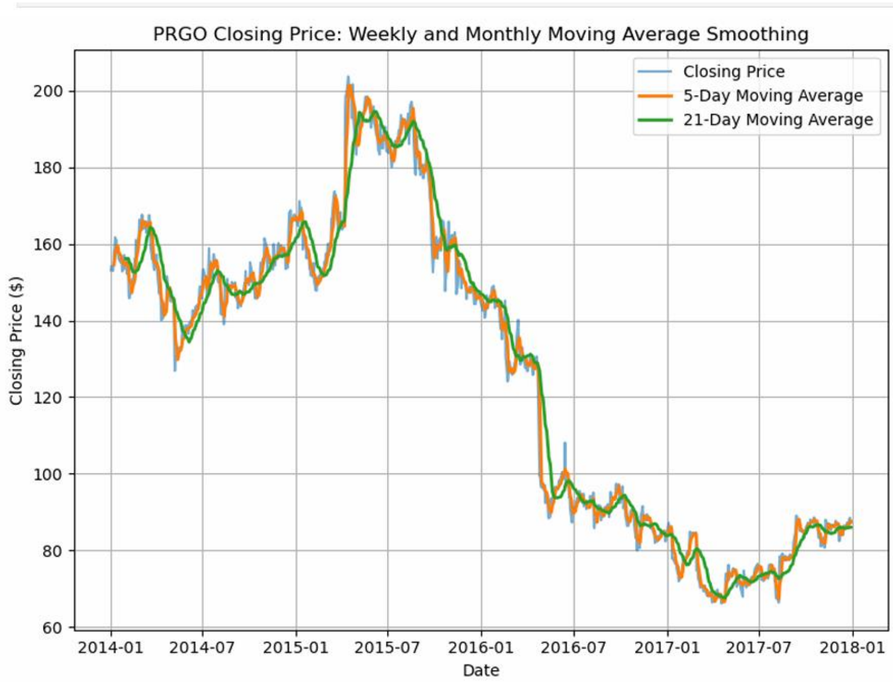

Figure 7. PRGO closing price with 5-day and 21-day moving averages.

For PRGO, the 5-day moving average tracks the daily closing price relatively closely and captures short-term turning points while filtering some of the daily noise. The 21-day moving average is smoother and lags the 5-day average around major changes, making the longer-term direction easier to see. The moving averages clearly reflect PRGO's transition from the 2014–2015 growth phase into the sharp decline beginning around late 2015. During 2016 and much of 2017, both averages remain substantially below the earlier peak, while the later period shows flattening and a modest recovery.

The moving-average plot therefore supports the decomposition result: PRGO's dominant characteristic is a long-term downward trajectory rather than a recurring seasonal pattern. The 21-day average is particularly useful here because it suppresses much of the short-term noise and makes the prolonged decline more apparent.

8. NVDA versus PRGO: Comparative Interpretation

Dimension	NVDA	PRGO
Cumulative growth	Strongest; very large positive growth	Weakest; losses over the study period
Daily volatility	Highest: 2.37%	High: 2.14%
Trend	Strong and accelerating upward trend	Early growth followed by prolonged decline
Seasonality	Present but relatively small	More irregular but relatively small
Residuals	Notable irregular spikes	Irregular movements and breaks
5-day MA	Tracks short-term movements closely	Tracks short-term fluctuations and turning points
21-day MA	Smooth view of medium-term upward direction	Smooth view of prolonged decline and later stabilization

NVDA and PRGO provide a useful contrast because they occupy opposite ends of the cumulative-growth ranking. NVDA combines exceptional growth with the highest measured daily volatility in the sample. PRGO combines negative/weak cumulative performance with comparatively high volatility. Their decomposition and smoothing plots both point to trend as the dominant feature, but the direction of that trend is fundamentally different.

9. Key Findings

- The full dataset contains 497,472 observations across 505 stock symbols.
- AAL, PRGO, NVDA, NUE and NTRS each had 1,007 observations and were selected for the comparative stage.
- NVDA recorded the strongest cumulative growth among the selected stocks.
- PRGO recorded the weakest cumulative growth and incurred losses over the study period.

- NVDA had the highest daily-return volatility (2.37%), while NTRS had the lowest (1.38%).
- Seasonal decomposition using a 252-trading-day period indicated that seasonality was relatively small for both NVDA and PRGO compared with their long-term trends.
- NVDA's dominant pattern was sustained and accelerating growth, especially from 2016 onward.
- PRGO's dominant pattern was a rise into 2015, a prolonged decline from late 2015 through 2016, and subsequent flattening/stabilization.
- The 5-day moving average was more responsive to short-term price movements, while the 21-day moving average gave a smoother medium-term signal.
- Moving-average smoothing reinforced the decomposition results for both stocks.

10. Conclusion

The time-series analysis demonstrates two very different stock-price trajectories within the selected sample. NVDA experienced strong and sustained growth, particularly during the latter part of the study period, while PRGO experienced considerable short-term fluctuations followed by a prolonged decline after its 2015 peak.

Seasonal decomposition showed that the observed movements were dominated by long-term trends rather than the assumed annual seasonal cycle. Moving-average smoothing reached the same broad conclusion: the 21-day average made the underlying direction clearer by reducing daily noise, while the 5-day average retained greater responsiveness to short-term changes.

Taken together, the results show why multiple time-series techniques are useful when analysing financial data. Percentage growth provides a comparable performance measure, volatility quantifies short-term fluctuations, decomposition separates trend from seasonal and irregular components, and moving averages make underlying direction easier to interpret.

11. Limitations and Considerations

- The analysis is descriptive and historical; it does not constitute a forecast or investment recommendation.
- The seasonal decomposition uses an assumed 252-trading-day period. This identifies patterns under that specified cycle and should not be interpreted as proof of economically meaningful seasonality.
- Volatility is measured using the standard deviation of daily percentage returns and therefore captures one dimension of risk only.
- The comparison is based on five stocks selected because they had the maximum number of observations, rather than on sector, market-capitalization or investment criteria.
- The supplied project solely focuses on closing prices; additional variables were not incorporated into the analysis.
- Moving averages are smoothing tools and are not, by themselves, sufficient evidence for a trading decision.

12. Technical Appendix

The analysis was conducted in Python using pandas, matplotlib, seaborn and statsmodels. The principal operations documented in the supplied analysis were:

1. Load the CSV dataset with pandas.
2. Inspect the data using head(), describe(), info(), unique() and value_counts().
3. Identify the five symbols with the maximum observation count.
4. Convert the date field to datetime and sort each selected series chronologically.
5. Calculate cumulative percentage growth from the first observed closing price.
6. Calculate daily percentage returns and their standard deviation.
7. Apply statsmodels seasonal_decompose(model="additive", period=252).
8. Calculate rolling means with window=5 and window=21 for moving-average smoothing.
9. Visualize the original series, decomposition components and moving-average series with matplotlib.